

ActInf Livestream #001.1 ~ “Narrative as active inference”

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Duration: 1:28:16 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello everyone welcome to team com podcast one tentatively titled really looking forward to this discussion i'm daniel and along with me are yvonne alex do you want to introduce yourself briefly yeah my name is alex i just can't say what i'm a researcher in system management school in moscow by now just for protocol hello my name is ivan i'm just interested in internet active inference stuff great well to be interested in active inference is all that's needed for this paper and this discussion so let's get right into it we're gonna structure the discussion like this first we'll start with some warm-up questions then we'll get to the paper itself and this isn't the end-all be-all format for paper discussions but a few big areas that we'll hit upon we'll ask what are the big questions are this paper what do the authors want to address then we'll go through the abstract line by line because it's really important to understand how the authors summarize their own work we'll look at the road map of the paper at which sections come in which order so that we can have a big bird's eye perspective on where the paper is going then we'll go down into some definitions and claims that the authors make we can ask whether we agree with their definitions or whether their claims resonate with how we think about the world then we'll have time to have any questions that we want to that are about the paper and then also there's a few slides at the end where we prepared some visualizations about markov blankets and about active inference itself so let's get to the warm-up questions and so either of you can maybe give a first thought here what what is a narrative and how is a narrative different than a story and this can come from the free energy active inference perspective or not but one of you two guys go ahead uh at the moment uh i believe what first we what we need to do what we need to define with difference because uh if we have two words uh it would mean something different but as in any case it's depends on perspective and interest from how we see it and um [Music] maybe his uh narrative and ontology is also some different and ontology is different from the story as well so some kind of more formal definition for it we have a great review on different papers presented in the text and so it looks

like as state of the art in this field from scientific point of view but we if we go more engineering way and trying to find a way to and approaches to use functions of his narrative scientists and lately stories very nice and we'll get to see exactly their definition of narrative it's actually the first sentences of the paper uh yvonne anything there no cool another question that is sort of a fundamental in communication but also very important applied question is how individual and collective narratives interact so do groups genuinely have narratives or do group narratives just consist of individual narratives we can i mean we can move past these these are really just for warming up and thinking about what the paper is arising out of and so just the last warm-up question that anyone could like a stretch before hitting the paper before you run the paper you might want to stretch with these questions and just to to really get us thinking big about what this paper could be about and what science can tell us about narratives this question is what does a functional evolutionary or statistical perspective on narratives look like so instead of just approaching it from a psychological or from an anthropological perspective how can we take it deeper and connect it to other areas any last warm-up thoughts or we can just go right into the paper uh maybe yes and it's about introduction and first sentence in introduction um about narratives a report of real or imagined events and maybe i was not so close inside but maybe we need to define some definition and perspective about events what is it events for us at this moment and in in systems uh and events in uh active inference what could be here like a fundamental for us good question and we're going to return to their exact definition of narrative we'll have it up on a later slide so keep that thought around and let's just go into the paper so the paper we're going to be discussing is called narrative as active inference an integrative account of the functions of narratives and it was posted on psi archive which is a pre-print server the version we're looking at is from june 18 2020 and the first author's name is nabil buzagarin apologies for pronunciation and uh there are several other authors as well the paper really sets out to address a few big questions the first question uh is what does the concept of active inference offer us in the context of narrative or when discussing the concept of narrative that's the big question and then to take it down one more level they're going to specifically ask how does this active inference perspective on narrative address questions related to identity episodic memory future projection event segmentation which is actually i trimmed it but it's related to what you're discussing storytelling enculturation and more so let's get to the abstract um it's kind of simulating reading a paper you know i start from a title then we'll go through the abstract and then we'll go into the definitions next but let's just go bit by bit through the abstract so we can

really see what the authors think they're doing in this paper first sentence is basically saying that narratives are important and other people have recognized it the second sentence says that research has identified several functions of narratives that are important uh if anyone has any thoughts that they want to add on to one of these sentences or disagreement they can uh also use the raise hand feature in jitsi so on the bottom left the hand if you want to uh say something um the abstract is not too contentious in this paper this is the key signal phrase that's marking what they're actually doing in the paper they're going to characterize social and cognitive functions those are the plurality of functions of narratives in terms of the framework of active inference which we're going to talk a lot more about in detail what is active inference and then they say that active inference is depicting the fundamental tendency of organisms to adapt update and maintain inference about their environment that's sort of the beginnings of a definition of active inference which we're going to come back to they then review literature related to the function of narratives in all of those areas that we discussed identity event segmentation episodic memory etc and then the last sentence of the short abstract is we recast these functions of narratives in terms of active inference outlining a parsimonious model that can guide future developments in narrative theory research and clinical applications and so this is kind of a common theme that we see in active inference and free energy principle related research at this current moment in time the second half of the 2010s and the 2020 period which is um the authors usually consisting of a team with some disciplinary experts as well as some more free energy or active inference experts will approach an area like narratives or depression or evolution and they'll seek to integrate multiple phenomena under a common model so instead of saying well you got the short-term memory system and the long-term memory and the declarative memory and this you know this type of memory they'll sort of recast a lot of previous models into one common framework so that's what active inference can do and the idea would be that by having an integrated framework we're going to get multiple benefits the first is that we have an elegant and a parsimonious way for addressing the otherwise unrelated phenomena arising from the same physical system so we shouldn't have 10 models for describing 10 different kinds of memory in the brain we should have one model for memory in the brain and then different aspects of it could relate to these phenomena that we want to actually explain and the second thing so it gives internal coherence to the study of systems to use an integrated framework but also it provides external coherence because now the information or the thermodynamic components of the brain can be considered alongside something like a computer so that's the abstract here's the road map of the paper and i just

copied out the section titles and this is just so that even if some of these section titles sound a little bit like they're not fully specified or anything this is just so we can first sketch out the whole arc of the paper the first section is called the function of narrative and you can see they're going to be talking about narrative from a non-free energy non-active inference perspective they're going to be drawing on other research for example relating narrative to identity meaning making which is also sometimes called sense making and coherence as well as storytelling and they'll talk about the narrative practice hypothesis which is from social psychology the second section of the paper is explicitly about active inference where they define active inference and then they specifically relate how active inference can be thought of when looking at human social and cultural life and then the third section of the paper no surprise is narratives as active inference and so here is where they're going to combine section one and two into discussing how framing narrative as active gives us these features of a model that we'd like to see like it's elegant it's parsimonious it has high explanatory power it might lead to unique predictions unique explanations uh unique experiments that we could test these are all the things that we care about from a scientific and a meta scientific perspective so that's the big road map of the paper and now we can finally get into the first two sentences of the paper which are right here and alex this is where you were talking about the of narrative so it's great that they begin the paper just with the definition of narrative and i'll just read it out and then either of you or both of you i'd love to hear your thoughts where does this resonate with you where does it not resonate they write narratives are reports of real or imagined events which can be presented in language either verbally or textually or through sequences of images or other symbols that's their first short definition and the second sentence is the narrativity of these reports lies not just in their content which may point directly to specific events but also in their structure that is the ways in which the linguistic or other symbols convey a set of relationships among events especially temporal relationships so what do you guys think here for me what is important here that uh its present idea about uh ability uh in a language to present to represent maybe to a person firstly and he can share its uh in texts and describe himself and maybe try to understand somebody using these symbols yep yep yvonne anything no not right now sure so i think uh there's a few other words that we might want to also define and this is the nature of language right we could always go and ask well what is a report is the narrative only real if it's reported or what's the difference between real and imagined or what do you mean symbols so there's a lot of play in these definitions still but the features that i found interesting the first part is that narratives are specific reports of real or

imagined events so it's kind of like if a tree falls in a forest and no one hears it did it happen uh if no one reports the narrative there is no narrative okay if nobody's speaking there's no narrator so a narrative is a declarative event in that telling the narrative or experiencing it is what makes it real and then also this idea that a narrative is a sequence of symbols whether they're verbal symbols or text symbols these symbols are representative as symbols are and to bring it to the second section the string of symbols it's not just that there's a shannon entropy of the string or there's some surprise or there's some sequence it's actually that the representative nature of the symbols is inferring something about specific events again either real or imagined also how the events are linked and this is especially important in the context of things happening through time so that's their definition of narrative and pretty shortly after that we get to one of their fundamental claims and there's going to be a key sentence here that is actually meta scientific in the sense this is their evaluation of the research literature they say that narrative has socio-interactional and cognitive representational functions however the extant literature tends to focus on only one of these two aspects so in other words they're saying that the narrative has socio-interactional roles for example the narrative of a country is often manifested through politics or through politics uh or through policy and so there are social implications for the interactions between different narratives and then that sort of bodies interacting with each other but then inside of the head there's also this cognitive representational component of narrative and that has to do with how different people experience narratives or how different altered states of consciousness influence the ability to generate narrative and then they write in this paper we propose a unified model that accounts for both of the functions of narrative so this is where they're taking stock of the previous state of the literature they're saying look narrative literature falls into basically either one of two categories either it's about treating the person as a black box and then looking at how narratives occur in the social context or it goes the other approach which is it zooms in on the brain or on the mind from a neuro or from a psychological and that ends up giving not enough emphasis to the social components of narrative so my questions for you two are what are the implications of this unresolved dichotomy and what might an answer to this problem look like in other words what would it look like to have a satisfying science of narratives and what are the implications of us not having that satisfying science uh if there will be an implication for one common model and it will be possible to predict outcomes for testing this model in different applications this could be more scientifically or at least engineering yep and another engineering perspective on this would be false positive and false negative so let's just say that

we have two different methods for dealing with narrative one of them doesn't give enough emphasis to what's happening inside of the head that's the socio-interactional one and then the other perspective on narrative doesn't give enough emphasis on what's happening outside of the head and so we're going to end up with false positives and false negatives we're going to end up with lost in translation we're going to end up with situations where one approach leads to one prediction but then a different approach leads to a different prediction and without some sort of higher level theory to help us resolve that there's just not going to be clarity so we can see that having this unresolved dichotomy not just is it theoretically unsatisfying uh it'd be like if you had one model for physiology that happened you know just in the liver and one and outside of the liver but uh that would be theoretically unsatisfying but also it might lead to some very real consequences when models don't agree and then people end up acting off of the bad information so no surprise here but they're going to resolve this dichotomy with active inference so here's how they define active inference and then we're going to spend a lot of time talking about active inference and adding in some more figures because there aren't any figures in this paper which is totally fine um but the crux of the issue is about applying active inference to the idea of narrative so we really want to understand what active inference is and then we'll use the active inference framework to potentially reduce our uncertainty about what narratives are so they write active inference is a new theory that provides an account of the nature and mechanisms of knowledge driven context appropriate action originally proposed as a theory of the structure function and dynamics of the brain in the 2010 first in paper active inference provides a broad framework that has been extended to explain living organisms and cognitive systems across multiple scales so that's the definition of active inference and i i just tried a little first stab there's two big key terms and they come up again and again in this discussion and in others and that's active inference which is i put it in idea cloud because this is the big idea that we're talking about and on the bottom right there is the free energy principle and the free energy principle is the background it's the uh it's the model that we're working under that we get active inference from and in the 2018 interview where Friston says that the free energy principle is beyond falsification it's at the axiomatic level so that doesn't mean it's indubitably correct in fact axioms exist beyond the space of correct versus incorrect because correct and incorrect only exist within the context of an axiomatic set and first compares free energy principle to something more like the principle of least action from physics in the sense that other patterns are fit within that so it's kind of like the it's the canvas that we're going to be drawing on and so here are a few of the key things that combine to give

us active inference one of them informational thermodynamics another topic is informational foraging and the that's the active part is the actual foraging for information then another term that comes into play is deep models so temporally deep models with a deep past and a deep future predictive component as well as you could almost call them thick models so models that include multiple modalities or multiple perspectives there's an element of the integration of ecology evolution and development so we're not looking to make a science of narratives that only applies to tweets or only applies to novels we want something that's fully grounded in ecology evolution and development and then also there's this key term the markov blanket which we're going to go into a little bit more detail in the coming slides but first off what do you think about that what is that uh what does that make you guys think about great overview about domains and basic stones for his ideas so really as free energy principle the gave us opportunity freedom to try to implement it to explain different uh systems and organisms so and active inference as a more focused on his domain around the brain about how its work on cognitive and metacognitive levels and so on yep and that's the key difference between the axiomatic principles which cannot be falsified in other words there's no observation that would falsify the free energy principle whereas evolution by natural selection just to give an example uh it could be if we found some sort of uh anomalous explanation or an anomalous observation then it could reduce the likelihood that a certain model is true in other words specific hypotheses arise from these other fields that i have written in the black or the dashed line like if you find that the organism is only looking one second into the future it's not a deep model so you can test whether it's a deep model or not you can test the role of ecology evolution development etc you could look at whether the actions of the organism are consistent with thermodynamics you know if more energy is coming out of the system than coming in then it violates thermodynamics so that's something specific and on the other hand free energy principle is a little bit different and uh there's so much to this and you know every every podcast we do every discussion we have we could spend so long discussing the fundamentals and so i feel like it's a good approach each time it's like a coat of paint we're just to keep on coming over the same topics again and again and going into them a little bit more deeply each time so let's look a little bit at the markov blanket and here i'm drawing from a paper of gallagher and allen 2016 which is in the synthesis the journal and this paper is pretty nice because it's not by firstin uh it's a rarity you know but it's not by fristen but it contextualizes the work of fristen and collaborators from a philosophical perspective so here are philosophers trying to talk about markhaw blankets they're focusing on this center node a

and they write the circle that's shaded in gray the big one represents the markov blanket of node a this consists of its children parents and parents of children so here is how i'm thinking about this pedigree like a family tree is a metaphor for the generative process that leads to people or leads to a person you know that any person has to have two parents and if any children there has to be a partner so that's like s in this the parent of child so now what what's interesting to note is in a world where the rules are different for example if three people were required to create a child or we were able to create a cloning technology where only one person was required to create another human then the rules of this system would be different and so the mark called blanket would be drawn differently so generalize this idea of a structural form or a scheme a schema for causal forces so it's like the two parents cause a now this is not saying oh those are the only two things that cause a person to exist of course a civilization is required and all these other things but just within the context of a the all the only information you need to know is the parents and then there's times where these connections that are being extracted are statistical and there's times where they're causal and this is sort of um an interesting point because it really gets at the heart of what do these lines represent and so there's times where the real world system we want we don't know about where the edges are and we want to use observed data to get the edges so we have a bunch of stock prices and we want to run a model that says okay when gold goes up then we want an arrow from the gold node to some other node for when the price responds in relationship to gold so that's like a statistical or numerical extraction of these edges and that is uh related to approaches like structural equation modeling sem modeling or causal entropic uh type of causal entropic forces and there's other times like in the case of this graph where the relationships are genuinely causal so that's where the edge is actually representing a cause effect relationship of the states of the world so i'm going to nuance this marcotte blanket drawing a little bit so here it says in small script the subpartition of internal and external states is going to occur according to the free energy principle this is the caption of their figure and then i just added extra colors instead of just having the subscript so i blacked out the hidden external states to really emphasize that the things that are outside of the markov blankets are truly speculative causes so in this case it's like we don't know who the grandparents of a are we know that something led to the parents but we truly don't know what they are and then we know that something has to lead to the parents of the child but again we don't know where they are and then there could be downstream effects of the child but we don't know what they are and so a like the focal individual what's happening inside of that green node is uh the

internal states of a the system and focus the system of interest and then you can think of the things that a causes whether by itself or with a collaborator are like the active the active states those are the emitted actions of a and then everything that is coming into the markov blanket but isn't an internal state and isn't an active state is a sensory state and it's kind of weird to think about a pedigree as having sensory states but we'll see a few other versions of marcotte blankets where the term sense is going to be more appropriate but the point is whether you call it a sensory state or just a markov blanket input it's the same structure any questions on that or there's some more visualizations of markup blankets yeah maybe let's um see some more cool cool so another key point about marco blankets is that they can be multi-scale and so here i've borrowed a figure from ramstead at all 2018 and this is the answering schrodinger's question paper which is a extremely powerful multi-level phrasing of the free energy principle and so i've carried over those same hidden internal active and sensory states idea and then added this uh pink or magenta color to the outside and that's the markov blanket so we can see the hidden states are outside of the blanket and then the active states the blue circles are what penetrate outward those are actions that leave the monaco blanket and then the red states are in incoming sensory data now the reason we can talk about multi-scale markov blankets is reflected here so this whole thing let's just say this is like a cell you know inside the cell you have some stuff going on and then you have actions coming out and you have sense coming in but that cell is actually also in the context of a larger uh unit let's say a tissue and in this case a bunch of cells are next to each other and so each cell is doing this at the cellular receptor level but the organ can also be thought of as internal external active and sensory and what the mathematical claim is is that there's a continuity between these two different levels of analysis and so there's a few ways you can think about this continuity one is just the compositionality argument which is like okay if tissues are composed of cells and cells are composed of organelles it's sort of a markov blankets all the way down approach but there's also some formal continuities as well as some possible ways which we can make a model of this level which is a mixture of martial blanket states and then here at the uh more granular level we can actually parameterize those relationships and another way that we can look at the mark call blanket again i'm invoking the um previous magenta uh uh pedigree version so here's the pedigree version of the markov blanket around a the focal individual and then here i've used the same colors on a neural uh focused markov so here the markov blanket is around the brain or the brain of the body and the internal states are what are happening inside the brain action states leave the brain and sensations enter the brain and then the hidden states are what are

happening outside of the brain and then i put this kind of arrow in quotation marks and this is really where we get to the idea of a causal generative model of the world and that's when the brain speculates about how causes in the world might actually be related and it turns out that having a causal model of the world is one of the most effective ways to reduce your uncertainty because if you're always just waiting for sensations to show up and then trying to deal with them by changing your model or by doing actions it's going to be a very confusing world but if you can have a causal model of the world that says for example oh a car is something that emits a certain light and sound and heat and so none of those different stimuli are surprising because i have a generative model of the world where a car is driving in the street in front of me and so that's what explains the sensations you don't need to explain the sensations of the sound by changing your role model uh or by doing action per se just making this uh causal link in the world can reduce your uncertainty a lot but that's something that whether or not it's in the world it's actually being inferred by the brain and then just to unpack the math one more level and also connect the brain out to other systems here's a very similar drawing and this is just showing how um again the internal states of the brain influence the active states which influence the world plus the world has its own dynamics those relate to sensory breaths which influence the brain's internal model and again that plays back into action and similarly in the context of a non-brain cognitive entity like a bacillus we see that this internal model there's the exact same relationship the internal model influences action action influences the world plus it influences itself then the world influences observed states sensory states and those feed back into the internal model of the bacterium and so this is something that goes beyond brains or neurons or neuroscience this is something that has to do with and markov blankets any thoughts on that one uh i don't know maybe it will be late important i think about how markov blanket set up borders with outside of the system yeah we can consider and work with it as a real body and to define information flows or something like this yeah yeah so i think there we could go to section 1.1 of the paper narratives as cognitive schemas um maybe i'm sort of jumping ahead by just already talking about the active inference perspective but when i think about uh narrative and markov blankets one thought is like a joke uh you know three people walk into a bar it's like it's already told you the who and the what it's told you who is entering what the markov blanket you know we're not talking about the street outside the bar we're not talking about the uh office building down the road we're talking about the physical border of the bar and we're talking about three people entering that so we still there's a lot of uncertainty in that narrative but just by saying three people walk

into a bar um if the next line of the story is and then the chicken says it's like it's a non-sequitur because that is something happening uh in the but it's not part of the actual uh narrative it's not part of what was set up and so short stories or jokes or other narrative retellings are like little model systems where we can explore how exactly are narratives set up what are the inputs what are the transformations that happen during the narrative and then what are the outputs what do you think about that yeah i think it's possible to consider this perspective and um maybe one thing i want to add on this slide about sensations and sensations and action uh so if we can uh speak about mental action and a sensation like mental representation of person for himself for example yep yeah so in this way uh mental action is it the same action or uh you need to do a section with here you need to do a section with uh out of what yeah for sure great great great question and uh this is kind of for example when people talk about speech as a motor behavior then it's like it all works for when people are speaking but then what about the mental action of preparing speech or of hearing voices but not vocalizing them so there's this question about representation which is basically like you said is mental action action and then the flip side of that is is mental representation sensation is that fair yeah correct all right so here's what i would say to that um what we're experiencing as conscious entities i mean not speak for anyone else but what we're experiencing is always our internal generative model so that's why when we look out at the world we don't see a blind spot it's why the world appears to have similar clarity across the whole visual field have color across the visual field etc because we're experiencing our generative model and so when there's a uh mental uh arising when there's awareness of some phenomena that isn't necessarily driven by a change in the external conditions so you're in the room it's a dark room or it's a you know it's an unchanging light and all of a sudden you think about being at the lake so that's an experience that you're having um it could be a memory it could be a fantasy you could put different words on it but no sensory data entered you to cause that thought to occur and so in this model of the markov blanket the sensations are actually very very specifically the sensory information that's being transmitted through the primary senses so the internal states the dynamics of the internal states it can be like it's like wow i'm bored in this room i wish i were at the lake now i'm experiencing the lake and so that's you can still have that awareness and the representation of the lake but that's actually not happening through the sense uh thing so again just there we're always experiencing the internal generative model uh so experiences are not here experiences are out here this is purely about actual sense data and then in the exact same way action is actually referring to um what is physically embodied by

the organism and so if the organism is planning which strategy it's going to use policy planning and it's like okay i'm going to wait 10 seconds and then as soon as he goes left i'm going to go right i'm going to run past him that's all part of the general model of the it's part of the internal state and not until the organism actually acts is the action real so mental experiences and mental processes are all happening on the green and the red is literally smell taste sound vision and the blue is quite literally motor behavior does that categorize it or what do you think about that yeah cool inputs for me right now it's more clear good so um a few let's go to section three narratives as since um let's we can kind of get to the real crux which is what does it mean to take an active inference perspective and so one part that i thought that was pretty nice point was that this the second sentence of section three they say we suggest that the capacity to build and share narratives evolved so key it evolved largely because it provides group members with the ability to engage in cooperative action by avoiding unexpected states of uncoordinated or contextually inappropriate action eg this is how we do things here not in other ways and so you reduce uncertainty for other people in a cultural context when you drive on the correct side of the road it would lead to a lot of uncertainty if every single time people had to come to a sort of just-in-time decision about which side of the road to drive on and so norms whether they're implicit or explicit norms are really interesting to think in the sense of reducing uncertainty and so you can reduce uncertainty about somebody's um behavior in various ways and they suggest that this leads to an account of the conservative aspects of cultural practice or the traditional components really just the idea that anything is preserved is conserved or held as traditional from the past and that ensures that human agents that are in culture that way are able to reduce their uncertainty when working with each other what do you think about that or how does that seem to you too uh for me it's interesting to address from ideas of niche construction and really from functional side it's how you describe it it's reducing uncertainty and provides ways to be more effective in communication and our kind of stuff doing in collective way yeah so that's that's one on page 19. that's just one nice uh idea about how this reducing uncertainty perspective of free energy principle rather than maximize reward it contextualizes a lot of traditions so for example it's hard to understand how some cultural traditions or cultural norms are simply about maximizing reward because it's like reward for whom or how is it calculated or how is it implemented but rather when we switch out the thing that has to be done by culture as not just maximizing success or reward but actually just maximizing um the precision of the action it makes a lot more sense so then at the end of page 19 this is a second uh area where narratives and active inference are gonna

intersect they write coherent self narratives provide individuals with a mean to maintain a stable course of action pertaining to own particular goals and experiences which limits the occurrence of unexpected states and so that first account of uh cultural uncertainty reduction is something that's happening at the level of like a population or a group and this second uh sense of narrative is more like something that somebody tells themselves as a means to maintain a stable course of action so for example i'm gonna go ride my bike to get some batteries because my flashlight doesn't have any batteries and i'm riding to the store and then i get a flat tire so now i need to walk my bike to the bike store but after i go to the bike store then i'll be able to get those batteries so i can go to the flashlight it's like it makes sense because that story about i that story about what has to get done it's very flexible but it also prevents the occurrence of unexpected states and when unexpected states do occur like getting a flat tire it can be incorporated into my narrative about my day instead of causing me to freeze or halt like a computer if there's an unexpected exception might just freeze or crash but informational foragers real evolutionary creatures can't do that so that's one other very very personal uh way that narrative matters and then just on page 20 furthermore narratives are potential generators of innovation and change and this is sort of combining those previous two the collective narrative precision uh increasing components and then the individual precision increasing components of narrative and then this is sort of bringing them together and saying that by doing active like search of imagining of exploring imagining new possibilities in response to or anticipation of adaptation can reduce or uncertainty and so for example if we say we're going to reduce our uncertainty about the future of the climate by doing this and that kind of modeling and even though this or that weather pattern hasn't actually occurred we should build a new wall here and we should prepare this field for growing this crop so that's the way in which using predictive action um or acting in anticipation of environmental changes can reduce your uncertainty and that's a story that you're telling about the future it's a narrative um any other thoughts there um i want to say is for self narratives actually people spend most of their time living in their self narratives and as for really feeling themselves the goods they always creating some kind of unreal stories and uh really going sometimes going in the wrong way mentally yeah it's a one of the the tenants about what leads to an organism surviving is the organism has to have a an optimistic uh future model like it the organism has to have a world view where it sees itself existing in the future i mean why sleep if there's no tomorrow or why eat if there's no tomorrow or you know all these things and we see that kind of survival reflex playing out when somebody's being held under water their body will struggle because it's not

part of the expected sensory inputs to be drowning so on one hand we see this fundamental struggle for life which is deeply optimistic but then at the same time that goes awry when people sort of uh end up licensing themselves to be the judge jury and executioner so to speak of the world and then they're the star in their movie and they're walking around the world and everything is only mattering to the extent it makes sense to them or it benefits them so having this uh personal level is uh sometimes at odds with the collective narrative that's really a huge area of tension and makes me think a lot about this multi-level markov blanket you know here here's us you know here's one person let's just say on the left side here's stories coming in narratives coming in memes coming out so information in information out and the hidden model is someone's political beliefs or their beliefs about lord of the rings and then that person is actually taking part in larger systems and so here's the whole country or the whole group chat or the whole team and then that team also gets inputs and gives outputs and so that's coming back to your original question about what are markov blankets in the context of narratives and one element is markov blankets are defined by what comes in and what comes out and that's informational in this case and another answer or another aspect of the answer is because these markov blankets are also defined by their internal mechanics or at least their internal dynamics so dynamics would be referring to the patterns that are just simply observable and the mechanics would be the actual rules that lead to those dynamics arising we can think what are those rules and one example about where the rules um could be a little different let's just go back to that joke about three people in the bar we go three aliens walk into a bar on a where gravity is reversed so it's like i've put this extra rule into the narrative and now other stuff is going to happen in the narrative that locally is going to make sense because i've also specified this other mechanism that's going to be relevant but without specifying that uh that mechanism then uh it's assumed that the rules of gravity are regular and actually another funny thing and there's been some work on this is with humor and optimal surprise like simply to be insulting is not funny as well as to say what everyone else says is also usually not funny but it's on that edge of chaos where comedy and uncertainty and people some are laughing some are crying some people are trying to stop themselves from laughing all these things happen due to surprise to think about this joke and again not to get too over analyzing of jokes but when the joke ends with a twist and it was like you know uh something that wasn't necessarily specified the beginning but there's a there's a double entendre there's a pun these things are all humorous and what's humorous about them is they deliver that optimal level of novelty or surprise so let's go to section 3.1 cognitive schema and active inference on

page 20. and so this is going to be drilling down a little bit into the cognitive schema so this is internal to someone's brain these are the uh the cognitive patterns that one person has on board that are related to narratives has to do with active inference and so they basically write about how having specific narratives of events like i was going to the store and then this happened and then this happened or i went to college and then this happened those events organize your life they org they give meaning to what you have to do when you have to wake up and not just like a you know theological meaning like literally it just gives sense to your day it tells you which direction to head out of the house and then secondly there's this meaningfulness that arises from uh general events and collective and they basically say that what we can do and also i'll just point out that those two levels of analysis the cognitive and the collective those are referring back to when they said that the issue with the field is that the literature on narrative focuses on either one or the other so psychologists will focus on the internal cognitive schema of narrative and the anthropologist well focused or the cultural researcher will focus on the collective um and they're going to try to use active inference to bridge that gap and so basically they say on the quote that goes from 20 to 21 um what explains these associations why these individual collective narrative are adaptive in an evolutionary sense is basically because they facilitate the process of active inference in page 21. that is the capacity of narratives to represent the specificity of events and to represent multiple events in a meaningful and coherent way helps individuals to more accurately predict future events and therefore respond to changing circumstances in their lives so that's pretty key what do you think about that um as it's stated about active inference that here in paper but it's first of all adaptive behavior and for sure this part is about how it's work in real life for example and people are rarely act like this always living in some stores and describe for them for themselves future where future possible events and in terms for example engineering again you first need to create a description of future system and how it will work in future so it's way yep yes and um and just at the end of section on the bottom of 21 where they say in short having a narrative or a repertoire of narratives furnishes inference with a simple and efficient way to account for sensory evidence at hand and what that allows uh organisms to do is to then provide hypotheses that enable inference to the best prediction so you're telling the story and it's like okay so there i was i had the flashlight and the batteries and my bike was working and then you can actually make hypotheses even if you knew very little about the person you could then say well what do you think would happen if they found out that the batteries were dead oh they might go back to the store because remember they needed the batteries for the

flashlight or what would happen if this other thing happened so the narrative allows temporally related events to um become meaningful and in the context of artificial intelligence and in big data this is so important because the data overload means that let's just say that you had somebody who was they had agreed uh consented to provide you all their data so in the whole day you have their full movement patterns all their their smartphone and every web search and all their transactions all this stuff you still might never be able to reduce it to so simple a story in just a few sentences as i had to get batteries for my flashlight but then my bike tire got a flat that is really really hard to extract from just the gps data and the transaction data um because it's intermingled with so many other different components happening and so narratives are how we relate that sequence of events to ourself it's how we relate it to other people and then the challenge is to generate computational systems that can interact at that level of analysis um like you know okay google i want to you know get a haircut within three miles of where i am or something like that that's what we don't have and that's why the artificial intelligence community is so interested in these topics because potentially an act of inference based artificial intelligence uh both confusingly ai could have a more uh effective role in some of these questions so let's go three one one narrative identity and in this section they're gonna go a little bit more into that first person narration component and on 22 they write coherent narratives link together aspects of our experience in ways that decrease contradictions amongst choices values and goals and so the contextualization of narrative is well it was really important to me to um spend time with my family but then for work i had to go away for two days and so that narrative is what allows that person to have two things uh that are otherwise contradictory which is well if families are important and work is important then what are you gonna do but the way that that's narrated it ends up not being that big of a deal it isn't that big of a deal because as they point out coherent narratives can reduce the dissonance amongst the complex and overarching expectations we have about our lives especially with relationship to future so that's kind of i think one very interesting uh topic anything there or yvonne any questions or thoughts uh and again again i want to put about personal and uh fun functionally for coherent narratives that people in their in their lives uh usually not action we're trying to create a new narratives way why we don't uh do any action and uh when we speak about action it's about learning and learning it's about updating your model and it's looks like it's quite difficult for people stop to create new narratives just not going to do some actions yep very very true um two quotes that that really makes me think about on 22 about two-thirds the way down says self and identity narratives are framed against

a background of cultural norms and values so they also can reference alternative narratives that provide explanations for non-normative unusual inconsistent or untoward events in the individual's life trajectory so that'd be something like you know when i was growing up everybody uh was drafted into the army but then because of this situation i didn't have to go so that explains why the individual was doing something explains why they had the sensory experience of being at home for a specific interval of time but it also juxtaposes it against a collective potentially normative idea and so this has a lot to do with humans seeking to differentiate themselves from each other is one tendency we want to be unique in some ways but also there's other areas where uh we want to fit in and so there might be cultures where it's valued more to be fitting in or not there might be different dimensions where we fit in different dimensions where we don't that's kind of like the idea of intersectionality and then on the page 23 where they're talking meaning making and sense making and i i totally agree with that what you're with most people being in a non-active state the sensations coming in the you know media coming in or the internet coming it's just updating their internal states updating updating updating but it doesn't seem to be reflected that heavily in action sometimes and then they write on the middle 23 the meaning making process is an adaptive response to the occurrence of unexpected circumstances when expectations are broken an imaginative reorganization of our representations of ourselves and our environment is needed um but unfortunately if somebody sometimes gets a statistic or a fact that is not consistent with their worldview then that's where we see logical fallacies like oh ad hominem or that that person's just trolling or they're just uh you know trying to frustrate me or something because it's uh it's always possible to include into your view just the idea that that one fact is wrong or irrelevant rather than actually move into this totally new worldview because shifting the deeper attendance of your is uh anxiety producing and so then also just to what you said about niche they talk about how uh before a new sense of familiarity and coherence i.e a new niche that's like a narrative niche can be reinstated exploration of new meaning epistemic foraging must occur in order to generate tests and discover new adaptive inferences so that's kind of a cool idea let's go to uh 3.12 event narratives so this uh section 3.12 which is a good length uh we can kind of just skip to the summary on 27 of this section where it says event narratives are ideally suited for active inference because of their ability to represent past and future events respectively in segmented event sequence and that provides a pool of uniquely structured information that helps predict future events and structures these inferences in the form of cohesive future events so um it it seems like a very basic point to build the paper towards

which is that narratives help us recognize where we are recognize who we are personally and culturally recognize where we've been and then make predictions about the future that hopefully will help us reduce uncertainty these points are not major but i would argue that it's actually quite um encouraging and powerful that these sorts of claims which usually would be the qualitative uh end point of some sort of psychological or cultural analysis of rather those are the starting points for us having a quantitative analysis potentially to analyze narratives um but that this section we've been kind of discussing uh throughout so section 3.13 integrating the functions of cognitive schema narratives as in as active inference so that's one of the final sections before they turned to storytelling um and uh that just is about what it says it's gonna be about any thoughts on that one like that's kind of related to what we've been discussing as well where they say on uh 27 on the bottom future projections and episodic memory can be understood as mental action or policy selection and perception so that kind of relates to the idea of mental action yeah yep and then just to kind of finish out the uh body of the paper before we return to a few more images and close out the section three two storytelling regimes of attention and deontic cues so uh regimes of attention are this idea that um you're paying attention to something you know when the child is in school and they're being trained to pay attention to different sorts of cues and what symbols mean this idea that attention and that attention as shaped by active uh as on 29 where there's a few citations to constant at all 2019 and constant at all that the narratives direct what is salient to human agents um so it'd be like you know three people walk into a bar and one of them has three legs it's like one of those statements is surprising the one about three people walking into a bar is not but the one about somebody having a different number of body parts is and so narratives allow us to very very easily draw on the tremendous amount of information that we have in our generative model of the world and then use that to discover anomalies in our observations and deontic is a word that means duty so related to what somebody uh must or should do and theref and so we can think about narratives for ourself and collective narratives as being deontic so when jfk you know said ask not what your country can do for you but what you can do for your country it's almost like instead of thinking about the united states of markham blanket and how that is going to provide something for you as well you might want to think about you as a markov blanket and how you as a personal narrative active inference machine might also be playing a role in this larger system uh maybe he was thinking about free energy maybe he wasn't hard to say and then they kind of closed the loop in that paragraph in the middle of 30 where they write um deontic cues guide the sensory motor patterns through which agents engage

adaptively their niches action thereby guiding the ways in which agents update their recognition model so for example when my alarm clock goes off and i go okay i have 30 minutes to take a shower get dressed bike to school because i have to be sitting in this lecture hall so that i can receive the sensory input of the lecturer so that i'll be able to update my model about organic chemistry so that when there's a final i'm able to reduce my uncertainty about what the answer is so that i'm able to get this grade that i want and so the deontic cue is like the post note that might be on my alarm clock that says get up you have to go to organic chemistry but that ends up guiding the sensory motor patterns and ideally that is a adaptive engagement through action and then that leads to differences in perception and learning and then this relates to a lot of other work um i think it might be the vessier uh at all 2020 citation on page 31 which is the idea of thinking through other minds which is a more recent development in the free energy slash active inference cultural so let's go just to their conclusion the authors are going to summarize what they've been discussing they write we have proposed an integrative model of the diverse functions of narrative as tools for active inference in ways that clarify their cognitive and social uses so again the big point of the paper was that there's this cognitive internal level and this social external level and the literature tends to focus on one or the other they brought the big guns out active inference and so they're going to integrate cognitive and social under the umbrella of active inference so that's going to give narrative and it's also going to give external coherence because now we can study narrative alongside other informational and thermodynamic processes then they in the following sentences recall a bunch of the specifics they talk about how narratives are useful for segmenting events for uh building an information pool for making future projections for meaning making and sense making for belief updating for assimilating unexpected information for maintaining cohesive integrity of our complex network of expectations so as to minimize internal contradictions as well as providing expectations to culturally common situations and typical life trajectories um and then they close by discussing that idea of the deontic narrative and the narratives that are about how the world could or should be and then that situates our role in the world and that actually shapes our sensory motor action patterns and that their last sentence is just that the importance and ubiquity of narratives uh the various domains of adaptation and sociality it should be explained not by appeal to entertainment or simply information exchange or communication only but rather by phrasing this all within the context of active inference we get a very very powerful framework to get internal and external coherence for narratives um then just sort of in closing i thought i would go to a paper from

march 31st to 2020 uh active inference on discrete state spaces a synthesis and just a little bit looking at how uh active inference what what could be the next steps and this is something uh you know the three of us have talked about quite a lot which is this paper is awesome it's also a qualitative perspective that doesn't dig into all the details that we wanted to see resolved that's why research is on the edge and so these are just a few of the ways in which active inference has been specified and so we might begin to think how narrative could play into some of so here we see that familiar diagram with the brain internal states action states external sensory states and then there's basically two things that are happening two optimization processes that are happening at the same time perception is the brain making sense of sensory data and action is the brain's selecting policy and what happens is the way that free energy is minimized given the sensory data two things can happen you can either make sense of the sensory data by changing your worldview or you can change your sensory data by engaging in action and so it turns out that by different transformations of these mathematical relationships that are specified exhaustively in this paper we pull out a few key terms so here's value this is the discounted or the expected utility it turns out that the value u^m expected value of an action or expected value of a stimulus relating to questions about optimal control and cybernetics as well these questions about value uh actually arise uh purely from thinking about uncertainty through time uncertainty through time is also related to uh surprise this is why on both of these we see this $\log p$ be a given m u^m and this negative f and f and then u^m also we're able to introduce the topics of entropy self-organization synergetics homeostasis and of course model evidence this is the key u^m uh term $pebm$ that's represented in all of these so those are a few interesting topics uh this is another visualization that talks about how active inference and markov blankets have very specific other downstream mathematical relationships again not too much to discuss at this point but these are just to put up and then this is the brain and here are the different variables in their represented visually so here are for example the generative model and how things are being influenced and how voter cortex is ending up select policy moving to the motor cortex moving out and then maybe we can just close on this slide which alex i think you put in where we have two agents or more and here for agent one here's their regime of attention this is their internal model what they're paying attention to what is salient for them and this is related to who they are if they don't speak the language they can pay attention to the sounds but they can't pay attention to the meaning if they don't know the area they can pay attention to the words but not really the connections agent one engages in action and that kind of dumps into the world and the world

is full of epistemic resources cues this is just what is and then the world uh ends up dealing out sensations to both of these agents and then it feeds back and so it's kind of like the conversation would be okay this person is listening they speak it dumps it into the text room or the the jitsi video chat that's sensed by agent n agent n pays attention acts that goes back into the chat room and so there's like this infinity here uh and that's thinking through other minds that's how narratives are shared this is really where we want to go is thinking at a really formal level as well as an applied level how can we bring this kind of clarity to action sensation and collaboration how can we bring that to the system's perspective the complexity yep perspective and um i want to add here we see a green zone about epistemic resources and silence and as we see from this paper narratives are really one of fundamental part of human well-being and developments and culture social and so on so and if so much functions in it we need to uh see closer to things like about epistemic side of his narrative and silent silence part of his narratives and trying to find ways to have some kind of evaluation and qualification for such sources or narratives as well agreed agreed and again with this idea of like there's the internal coherence and then the external coherence we can return to that joke it's like in the context of the joke you can make a mark of blanket for the joke where three people walk into a bar that's literally the three people walking into a barkov blanket um probably there's three people walking to a mark called blanket they get lost um but then zooming out there's also the cultural context that the joke occurs within uh the kind of context that makes uh three people walk into a bar now that's common it's an american or it's an english joke format you know this many people walk into a bar or knock knock who's there now that might be a culturally specific thing and so that joke is going to be more or less expected more or less funny to somebody who's in that cultural context and so the epistemic resources and the salience have to deal with the stimulus in question like the joke but also it has to be taken into account that because we're talking about agents that are paying attention the culture of the agent is going to dictate what they find salient there might be things that are in the world that are not salient because the uh individual doesn't have the narrative that allows them to retain that salience um any closing thoughts especially yvonne i just i just uh have remembered one russian joke that's congruent with the english three people walking the barriers the bill goes in the forest it's uh it's start with the beer goes in the forest it has a lot of different ants in this story exactly and some of these cultural stubs like these starters for jokes it allows uh the format for anything it's kind of like a http packet you know you have the header that just says i'm an internet packet that's like saying you know when you say a bear walks into a forest it's like i'm telling you a

joke but you can't just say i'm going to tell you a joke right now and so these are signals that are extremely rich with and just having a text-based natural language processing ai will totally miss out on that cultural context and then let's say you told like a variant of the joke you know um uh an ant walks into the forest it's not a non-sequitur but it's not really referencing a real ant walking into a forest and these are these overlapping levels of cultural jokes and meaning that are really pointing the way to how rich the study could be with active inference so how how do you think what the first thing that is uh people don't have people of different culture don't have to do not missing the point of communication how can what do they can what can they do first to reduce miscommunicating yeah good question and especially in a digital and international context it's so that there's reduction of uncertainty so i mean just really quickly a few things would be just at every possible opportunity where there's any question about uncertainty i feel like always getting the uh consent from both okay we were just talking about this topic right and so just getting used to agreeing with each other and getting used to uh altruistically or just politely exchanging information starts to train us to work well together even when the situation is very tense so that's one part is where we can be extremely accurate in communication we should be extremely accurate and then on the technological side i think it's important just to communicate simply instead of using think oh we'll just use video and use body language and make sure that you're really expressive with your hands uh yeah the tone of voice and the hand motions even if they come across with no lag and the person is watching your screen different cultures might unpack different hand motions or different facial expressions very differently and so i think that when there's differences in culture that lead to challenges in communication it's critical to really just focus in on what you're actually trying to communicate and make sure that there's as little uncertainty as possible about that and then the cultural stuff deal with it as it's needed and whenever it's needed to deal with it um absolutely it's important to deal with it because nothing can be more than helping the team become more effective more precise i think in digital it's more simple to um to make clear enough communication between email between two people because in when we when we are talking in real life we can't to add some hashtag to uh describe what what do we uh what do we mean yep we can use a rich multimodal features so we could be having this chat and someone could say i'm sorry i just could you spell out the city that you mentioned and then they can type it in the text box or somebody can share their screen and draw something out things that couldn't happen in person i i agree these are all really important and that relates to epistemic resources so when the people who you're dealing with are all in the same discipline

and all the same culture maybe the epistemic resources are very professional but when we're talking about broad spectrum communication potentially the epistemic resources and the salience cues would be like using a color scheme that's accessible for everybody and using icons to demonstrate where the settings are like having a gear in the bottom right of jitsi the three dots having a gear that says setting that's really helpful because it's kind of like a universal icon for the settings and then otherwise it could be a situation where someone doesn't know english and they can't find the settings so then how are they going to be able to find the language uh option because they can't even find the settings section but with the gear it's um it's a salience cue that goes beyond culture uh and so that is what helps people get along together and again going back to like what are the actual applications of this if we could how individuals in different cultures are going to perceive different narratives and what role they see themselves playing in the narratives it could really open the door to extremely interactive and authentic narratives by that don't have to be situated as being at odds with some sort of evil system and it's they're they're the righteous crusader fighting against everyone else who's dumb or the system's corrupt and they're just you know doing what they can all these narratives that that uh hit the individual versus the group or versus specific groups lead to adversarial game theory and so not not that there's never going to be competition in the world but potentially by having these protocols for interaction and for communication just like we have protocols for trade we could have protocols for information that are also healthy yeah cool and as for protocols uh and you said before about signals and i think if you speak about culture it's really mostly uh shared by narratives and uh if he's considered culture not like a global uh gentleman but a local so each team could have own culture and they define it by narratives and it creates a niche by that supporting by tools and the tools really could support such protocols where some meta information added to communication and communication process is also structured according to some models from active inferences we see totally agreed and i think on a closing note this has been our first podcast live stream discussion for team kong so it's a a landmark event for us we're using jitsi for live streaming to youtube we used google slides for the presentation the pre-print servers psi archive and the internet in general are what allowed us to make these presentations and um we can really just see this as happening as just another event we're observing this live stream come to an end it's part of our narrative of having met in uh you know in service of these ideas and with a lot of the same interests and so yeah this was really a great discussion and i'm looking forward continuing these discussions yeah thank you very much for your leading

discussion i think results are interesting and we will use it in future for sure all right i'm going to terminate the live stream now